

Full Marks: 50

Time: 2 hours

Answer 5 out of 9 questions.

5 × 10 = 50 Marks

- ✓ 1. Define linear and non-linear data structures.
Write an algorithm to convert an Infix Expression to an equivalent Postfix Expression.
Convert the following Arithmetic Infix Expression into its equivalent Postfix form:
 $(F + G) * C - ((A + B) - (D - E))$ (2+6+2)
- ✓ 2. What are Advantages and Disadvantages of using a Linked List over an Array?
How will you represent the polynomial $4x^3 - 10x^2 + 3$ using a Linked List?
Write an algorithm to delete an element from the beginning of a Doubly Linked List.
Write an algorithm to concatenate two existing circular lists. (2+2+2+4)
- ✓ 3. Write a program to implement Singly Linked List using templates. Also design functions for insertion, deletion and search of a number. (4+2+2+2)
4. What is a Priority Queue? What operations does it support? Explain in detail Heap Data Structure that is used to implement Priority Queue covering MinHeap. (2+2+6)
- ✓ 5. What is Recursion? Explain internal stack implementation using recursive factorial function. Explain linear, binary, and multiple recursions. (2+3+5)
6. What is a tree traversal? Write preorder, inorder, and post-order tree traversal algorithms for a Binary Tree, with illustrations. (1+3+3+3)
7. What is the precondition of performing binary search in an array? Write a non-recursive algorithm to search an element in an array using binary search method. What is the time complexity of it? What is the advantage of using binary search method over linear search method?
What is the best case time complexity of an insertion sort algorithm? When does it occur? (1+4+1+2) + (1+1)
8. How does an AVL tree differ from a Binary Search Tree?
Construct an AVL Tree by inserting the following set of nodes:
B, W, M, T, R, J, D, S, Q, K, V, P
State at each step balanced or unbalanced, if unbalanced, rotation type.
What is the maximum height of an AVL Tree with n number of nodes? (2+6+2)
- ✓ 9. What is Hash Data Structure? Explain in detail methods followed in Hash Collision Handling. (3+7)